LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	NACA series, Vortex theory, Magnus effect, Kutta Jakubowski law, Span wise flow, down Wash wing tip and trailing edge vortices
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	NACA series, Vortex theory, Magnus effect, Kutta Jakubowski law, Span wise flow, down Wash wing tip and trailing edge vortices. Methods for preventing span-wise flow.
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of NACA airfoil charts and identification of vortex generation points on wing models.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aerofoil section?
(b) Define the term 'span-wise flow' in the context of wing aerodynamics.
(c) State the condition required for steady flow at the trailing edge of a wing.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• NACA Airfoil Series Classification • Vortex Theory Fundamentals • Magnus Effect Principles • Kutta-Jakubowski Law Application	L & Ds	PPT, IB	15 min
(b)	• Span-wise Flow Causes • Down Wash and Wing Tip Vortices • Trailing Edge Vortex Formation • Methods to Prevent Span-wise Flow	L & Ds	PPT, PDF	20 min
(c)	• Impact of Vortices on Aircraft Performance • Induced Drag Relationship • Winglet Functionality • Summary of Aerodynamic Concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the Kutta-Jakubowski law relates to the separation point of airflow.
(b) Describe the mechanism by which wing tip vortices induce down wash.
(c) List two methods used to reduce span-wise flow and their effectiveness.

12. Home Assignment and Follow Up	Prepare a brief note comparing the NACA 4-digit and 5-digit series, highlighting their specific applications in modern aircraft design as a bridge to Drag analysis in LP 14.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Drag analysis, induced, parasite, form, skin friction, profile and total drag. D-V curve.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Sub-Lieutenant, P No. 12345	Senior Instructor, Lieutenant, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min